

MicroStrategy to Power BI Transformation

1. What this exercise is

You've been asked to spend a few focused days looking at a real, currently running transformation problem: moving enterprise reporting off MicroStrategy (MSTR) onto Power BI.

What we're trying to understand:

- How do you research a familiar/unfamiliar problem
- How you structure an approach to something which works as a baseline
- How you reason for trade-offs, constraints and effort
- How you translate a couple of similar examples into a proposed design

Note: There is no single "correct" architecture we're grading you against word-for-word. We are far more interested in the quality of your thinking and how you got there than in whether you land on any design.

2. Before you start: research the context yourself

This exercise sits inside a live engagement for Sainsbury's Information Factory – a UK grocery and general merchandise retailer – who are moving their reporting estate from MicroStrategy onto Microsoft Power BI / Fabric. We want you to spend some time researching this context before you design anything. Treat this as you would be in a real client engagement kickoff.

- Who is Sainsbury – scale, markets, and where data/analytics would plausibly matter to them operationally.
- Anything publicly discoverable about their technology direction – cloud platform, data platform, any known Microsoft or Azure alignment, any public statements on BI/analytics strategy.
- General industry context on why large enterprises move off MicroStrategy.
- A basic technical orientation on MicroStrategy vs. Power BI if you haven't worked with one or both before.

3. The core problem, in outline

MicroStrategy and Power BI are not variants of the same design. A handful of structural differences drive almost everything hard about this kind of migration:

Dimension	MicroStrategy	Power BI
Calculation objects	Metric is a global, project-level object reused across reports	DAX measure inside a semantic model, evaluated against filter context

Schema shape	Snowflaked logical tables and views layered over the warehouse	Star schema strongly preferred; in-memory or Direct Lake model
Grid semantics	Row-level arithmetic, consolidations and subtotals happen inside the grid engine	Filter-context evaluation; row-relative math needs calculation groups / visual calcs
Parameterization	Prompts can drive the generated query and restrict what a user sees	Slicers only filter context; sharing a model grants access to the whole model
Security	Row- and object-level security assigned per user or group	RLS roles with DAX predicates; object-level security via model tooling

4. Three sample reports to work through

To keep this task-driven rather than abstract, work through the three MicroStrategy reports described below as if they were handed to you at the start of a real migration wave. These are simplified, business-language descriptions – you do not need MicroStrategy access or tooling for this.

Sample A – “Regional Sales Performance” grid report

- A crosstab: Region and Product Category on rows, Month on columns, with Revenue, Margin % and Units Sold as metrics.
- Margin % has a threshold rule: green above 20%, amber 10–20%, red below 10%, shown as cell shading.
- Subtotals roll up by region; a grand total row sits at the bottom.

Sample B – “Store Operations” dossier

- A multi-page dossier: an overview page with KPI cards (Sales vs. Target, Footfall, Stock Availability) and a trend line per KPI.
- A Region selector sits above the page and drives every visualization on every page when changed (apply-to-all behavior).
- The second page contains a bar chart of sales by store, with a drill path from region → Store → Department.

Sample C – “Category Manager” report

- A grid showing Supplier, Product and Sales metrics, restricted so each category manager only ever sees rows for their own assigned product category – enforced by MicroStrategy security filters tied to the logged-in user.
- A prompt lets the user choose a date range before the report runs.
- A top 10 filter limits the grid to the ten best-selling products within whatever category the user can see.

5. What to submit

Produce a short deck (5–8 slides – this is about clarity of thinking, not volume) structured around five sections:

1. Problem statement – in your own words, tie together the business driver (from your research), the client context, and what's structurally hard about MSTR → PBI.
2. Solution framework – the stages you'd sequence the work into and why (e.g. how you'd decide what to automate vs. review vs. hand-build), without assuming unlimited time or budget.
3. Draft architecture – a sketch of the major components you'd stand up and how they talk to each other. A diagram plus a short paragraph per component is enough; this is not a full technical spec.
4. Report build – your proposed Power BI translation for Samples A, B and C above: which visual(s) and model changes you'd use for each, and why. This is the most “hands-on” part of the task.
5. Automation plan – what you'd lean on to move faster: an external/commercial migration tool, an internally built accelerator, or a hybrid – and your reasoning for that call.

Please also add a short closing note: where you personally would plug into a delivery pod for this kind of work, and what from your own background you think transfers well (and where it might not).

6. Logistics

Item	Detail
Prep time	4–5 working days, at a pace that fits around your other commitments
Format	A short slide deck or one-pager – not a working prototype or code
Live session	15–20-minute walkthrough of your approach, followed by 15–20 minutes of panel Q&A
What to bring	Your deck/one-pager, and be ready to talk through your reasoning – not just the output

Good luck – and remember, we're evaluating how you think, not whether you already know this platform pair.